Mateusz Płoszaj-Mazurek

Building researcher · Daylight, building regulation and computation

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Researcher at the Faculty of Architecture, Warsaw University of Technology. My PhD and the 2020 and 2024 articles in Energies developed machine-learning, parametric and BIM-based methods for low-carbon architectural design. My current work moves the same approach to daylight: fast surrogate models that predict climate-based daylight metrics from geometry, and parametric studies of how daylight requirements perform.

Publications

A workflow for LLM-assisted IFC data enrichment to support circular reuse of large-panel building systems in Poland
with A. M. Tofiluk · Journal of Building Engineering 120, 115397 (2026) · doi · open access

Green Data Centres: Sustainable Solutions with Green Energy and Green–Blue Infrastructure
M. Grochulska-Salak, E. Maciejewska, P. A. Bujak et al. · Energies 18(24), 6592 (2025) · doi

Artificial Intelligence and Digital Tools for Assisting Low-Carbon Architectural Design: Merging the Use of Machine Learning, Large Language Models, and Building Information Modeling for Life Cycle Assessment Tool Development
with E. Ryńska · Energies 17(12), 2997 (2024) · doi

Methods to Optimize Carbon Footprint of Buildings in Regenerative Architectural Design with the Use of Machine Learning, Convolutional Neural Network, and Parametric Design
with E. Ryńska and M. Grochulska-Salak · Energies 13(20), 5289 (2020) · doi **136 citations**

Machine Learning-Aided Architectural Design for Carbon Footprint Reduction
Builder 276 (7/2020), Builder Science, pp. 35–39 · doi

Digital methods of supporting architectural design process and the analysis of the carbon footprint of buildings
PhD thesis · Warsaw University of Technology (2022)

W kierunku architektury prośrodowiskowej – sztuczna inteligencja jako wsparcie architektonicznego procesu projektowego
Chapter in: Architektura i urbanistyka wobec sytuacji kryzysowych, eds K. Solarek, C. Głuszek · Oficyna Wydawnicza Politechniki Warszawskiej (2022), pp. 63–75

Whole Life Carbon Roadmap for Poland: How to decarbonise the built environment by 2050
A. Kuczera, M. Płoszaj-Mazurek, A. Hadziivanov et al. · PLGBC with WorldGBC and EBRD (2021), 52 pp. · doi

Budownictwo drewniane a architektura zrównoważona
Chapter in: VII Forum Holzbau Polska HBP2023, ed. M. Wnorowska · Forestor Communication (2023), pp. 215–222

Parametric Optimisation of Carbon Footprints
Chapter in: Regenerative Design in Digital Practice. A Handbook for the Built Environment, eds E. Naboni, L. Havinga · COST Action RESTORE (2019), pp. 258–263

Parametryczna optymalizacja śladu węglowego budynków
Chapter in: Miasto dla ludzi – miasto jutra, ed. B. Gronostajska · Politechnika Wrocławska (2019), pp. 303–315

Prescriptive and Performance Based Daylighting Regulation as an Architectural Experiment: Parametric Simulation of Polish Residential Standards
Architecture of Challenges: Experiment in Space – the Space of Experiment, ed. A. M. Wierzbicka · Warsaw University of Technology (2026), pp. 160–161

Implementing Building Energy Modelling in a Master's Level Architectural Design Course: Preliminary Insights from a Pilot Study
with M. Gołębiewski, M. Pierzchalski · Architecture of Challenges (2026), pp. 62–63

Predicting Daylight in Milliseconds: A Computational Machine Learning Surrogate Model for Early-Stage Urban Housing Design Simulation. Artificial Dataset Generation
Poster · Young Scientist Forum "200 Młodych na 200-lecie Politechniki Warszawskiej" (2026)

Artificial intelligence for reducing carbon footprint of buildings. A comprehensive approach using convolutional neural networks and large language models in architectural design optimization
Redefining Cities in View of Climatic Changes 2023: Abstracts · Warsaw University of Technology (2023), pp. 96–97

Artificial intelligence and machine learning for low carbon architecture. Prototype carbon estimation tool development case study
Structures and Architecture. A Viable Urban Perspective?, eds P. J. S. Cruz, M. Frier Hvejsel · CRC Press (2022), vol. 2, pp. 345–346 · doi

Parametric Optimization of Carbon Footprint
YRSB19: iiSBE Forum of Young Researchers in Sustainable Building, Prague · Czech Technical University (2019), pp. 165–174

Applying Machine Learning in Integrated Architectural Design for Life Cycle Carbon Footprint Reduction
Redefining Cities in View of Climatic Changes 2019 · Warsaw University of Technology (2019), pp. 117–118

Analytic radiometric descriptors enable geometry-flexible per-sensor daylight prediction without learned spatial encodings preprint
SSRN (2026) · code github.com/Curiosit/OmegaFlux · software doi:10.5281/zenodo.21781259

Akademik na Solcu, Warsaw
Project publication, with M. Czechowicz · Architektura-Murator 4/2025, pp. 54–64

Przedszkole pasywne, Reguły
Passive House kindergarten, project publication, with D. Sibińska, J. Sokołowski et al. · Architektura-Murator 5/2023, pp. 20–34

Prefabrication and sustainable design: a three-part series
with A. M. Tofiluk · Builder 4/2022, 7/2022 and 10/2022 · plus "Prefabrykacja w architekturze zrównoważonej", Zawód: Architekt 9–10/2022

Nowe wyzwanie dla polskiej architektury
J. Biernacka, P. Biernacki, P. Dowgiałło et al. · Architektura-Murator 3/2019, pp. 30–32

Selected projects

2025–26 — **OmegaFlux**
Per-point daylight prediction from geometric view descriptors with gradient-boosted trees, exported to ONNX and run in the browser. Dataset generation, held-out validation against Radiance, transfer to unseen room shapes. curiosit.github.io/OmegaFlux

2021 — **Whole Life Carbon Roadmap for Poland** PLGBC, WorldGBC, EBRD
Co-author, with Alicja Kuczera, of the national roadmap for decarbonising the built environment by 2050.

2018–now — **Multi-criteria building optimization**
Evolutionary and multi-objective optimization of energy, daylight and carbon in parametric models; taught as a course module.

Grants and awards

2024–25 — **Principal investigator, IDUB research grant** · Circular economy in architecture: reuse of prefabricated elements and digital design techniques · Warsaw University of Technology, CPR-IDUB/238/Z01/2024

2024 — **First prize, Best Mashup Project** · AEC Hackathon, Copenhagen

2024 — **Rector's team award for organisational achievements (first degree)** · Warsaw University of Technology, academic year 2022/23

2023 — **Rector's team award for teaching achievements (third degree)** · Warsaw University of Technology, academic year 2021/22

2023 — **Competition win: energy-efficient multi-storey housing in Poland** · Winning entry by Studio Widoki, with Bjerg Arkitektur as consultant · Carbon footprint calculations by Bjerg Arkitektur: prefabricated timber, about 6 kg CO_2e/m^2 per year over 50 years

2022 — **PLGBC Green Building Awards: best green building without certification** · Passive House kindergarten in Podkowa Leśna, Bjerg Arkitektur; award received on behalf of the practice. Also a distinction for an educational building.

2022 — **Distinction for the doctoral thesis** · Innowator Mazowsza competition

Talks and panels

2023 — **Pro-environmental design: training sessions for architects** · Saint-Gobain · two sessions, Jura Krakowsko-Częstochowska and Żnin, October 2023

2023 — **ESG Digital Leader Academy** · Proptech Foundation · development programme for managers implementing ESG strategy with digital tools

2023 — **Lublin on its way to Twin Transition: Danish–Polish workshop** · Lublin, 13 April 2023

2026 — **Discussion panels** · IV International Conference Architecture of Challenges: Experiment in Space, Warsaw, 29–30 June 2026

2025 — **Session "Architecture of Community" and discussion panels** · III International Conference Architecture of Challenges: New European Bauhaus – Building Community, Warsaw, 30 June – 1 July 2025

2025 — **Decarbonising buildings in the climate crisis: policy, practice and the investor's view** · PLGBC Green Building Summit

2024	**Digital and business challenges for developers in 2024: non-financial reporting and carbon footprint** · Building & Human Summit, Łódź, March 2024
2023	**How to choose software for managing a building's carbon footprint** · Webinar, Proptech Foundation, 13 December 2023
2023	**Energy resilience as a priority in urban and architectural design** · III Kongres Regeneracja Miast, Łódź, 31 May 2023
2023	**"Trochę architektura, trochę biologia"** · Młodzi do Łodzi, conference of Architektura-Murator, Łódź, 26–27 May 2023
2021	**Conversation with a Sweco sustainability and BIM lead** · Miasta Przyszłości, 4th edition, online
2021	**Launch of the Whole Life Carbon Roadmap for Poland** · Webinar with PLGBC, EBRD and WorldGBC, 9 June 2021
2018	**"Zero Carbon. Road to sustainable home"** · 8th Green Building Symposium, PLGBC
2025	**Sztuka vs. Sztuczna Inteligencja. Algorytm z duszą i emocjami – to tylko kwestia czasu?** · 22nd Warsaw Art Fair (Warszawskie Targi Sztuki)
2025–26	**Early-phase building carbon footprint optimization with Carbon Designer 3D** · Carbon Experts Summit, 2025 and 2026
2023	**Special presentation, "The role of steel in the future"** · V Dzień Stali, BUDUJ ZE STALI, Warsaw, 28 November 2023
2023	**Timber building and sustainable architecture** · VII Forum Holzbau Polska, Jachranka, 25–26 April 2023
2023	**Low-carbon sustainable architecture and the use of BIM** · Mistrzowie Krajobrazu 2023, online, 21 April 2023
2022	**Bjerg Arkitektur's approach to climate in construction** · Kongres Regeneracja Miast, Łódź, June 2022
2021	**How prefabrication lowers the carbon footprint of buildings** · Sztuka Architektury, online, 27 May 2021

Experience

2017–now

Researcher and lecturer

Faculty of Architecture, Warsaw University of Technology · doctoral candidate from 2017, academic staff since 2021

- Daylight regulation, simulation and surrogate-model research; see publications.
- Teaches Grasshopper and multi-criteria building optimization (400+ students so far); coordinates a lecture cycle on public buildings; runs a second-year design studio.
- Designs parametric experiments and evaluation frameworks: thousands of simulated room variants, held-out test splits, convergence checks.

2024–2026

Software engineer, interim product owner

One Click LCA, Helsinki · remote

- Built features for Carbon Designer 3D, an early-design tool for architects (TypeScript, React, Three.js, AG Grid).

2014–2024

Architect, Certified Passive House Designer

Bjerg Arkitektur · Hjørring 2014–16, Warsaw 2016–24

- Single- and multi-family housing, schools, kindergartens and offices, from 100 m^2 to 15,000 m^2.
- All design phases: concept, permit and construction documentation, and architectural supervision on site.
- 2022–2024: architecture, sustainability and software consultant, building in-house design tools; co-author of the Polish whole-life carbon roadmap.

Teaching

7 years

Introduction to Grasshopper

"Cyfrowe techniki geometryczne": parametric modelling and multi-criteria optimization of energy, daylight and carbon.

2020–

iBIM postgraduate programme (Interdisciplinary BIM Studies)

Introductory and advanced Grasshopper, environmental and daylight analysis with Rhino.Inside.Revit and Grasshopper, Python and C# scripting, for practising architects and engineers. Hybrid.

	2026–	**Sun, Daylight, and Architecture** English-language course on the role of the sun and daylight in architectural and urban design, co-taught with Krystian Kwieciński.
	2026–	**Digital design in sustainable practice and research** Elective seminar "Zastosowanie projektowania cyfrowego w zrównoważonej praktyce projektowej i badaniach naukowych".
	ongoing	**Computing Architecture for Sustainable Environment** Seminar for fourth- and fifth-year students, including daylight simulation.
	ongoing	**Design studio and lecture cycle** Second-year studio on small public architecture; coordinator of the semester-seven lectures on public buildings; English-language Solar Seminar.
	2021–	**Techniques Integration, ASK** Co-taught course integrating technical and environmental design.
Service	2026–	**Guest co-editor, *Buildings* (MDPI)** · Special issue on performance-responsive architecture and urban design
	2025–	**Reviewer, 21 reviews for 9 journals** · Buildings (10), Energy and Buildings (3), Architecture (2), Automation in Construction, Artificial Intelligence Review, Digital, Discover Artificial Intelligence, Energies, Scientific Reports
	2024–26	**Jury member** · National competition for the best pro-environmental diploma project; judge-rapporteur in 2025
	2023–26	**Conference committees** · Organising committee, Architecture of Challenges IV (2026); scientific and organising committees, Redefining Cities in View of Climatic Changes (2023)
	2018–23	**Editor, catalogue of the best diploma projects** · Faculty of Architecture, Warsaw University of Technology, five volumes, two co-edited with A. Grabowska
	current	**Auxiliary supervisor** · Doctoral candidate at Warsaw University of Technology
	current	**Member, Sustainable Architecture Circle, SARP**
Education	2022	**PhD in Architecture** Warsaw University of Technology Digital methods of supporting the architectural design process and the analysis of the carbon footprint of buildings: parametric optimization and machine-learning methods for design.
	2014–2016	**MSc in Architecture** Aalborg University, Denmark
	2023–2024	**Certifications** Master BIM Software Developer (That Open Company, 2024) · Harvard CS50x (2023) · Certified Passive House Designer
Skills	**Building domain**	Climate-based daylight (sDA, ASE, UDI, EN 17037), insolation, Polish and Danish building regulations, Passive House, building carbon footprint
	Languages	Polish (native), English (fluent), Danish (communicative), Chinese (basic)